Reg. No. | | | | | | | | | | |

[13CE65/12CE65]

SIXTH SEMESTER B.Tech./B.E. EXAMINATION, JUNE – 2016

DESIGN OF HYDRAULIC STRUCTURES

Time: 3 Hours

Maximum Marks:70

Instructions:

Answer any **FIVE** full Questions.

Use of Design charts is allowed

Assume any missing data suitably.

1.	a.	Explain multiple step design method for design of gravity dam.	10
	b.	What are the various types of Arch Dams?	04
2.	a.	Explain the stability analysis of down stream slope during steady seepage with neat sketch.	08
	b.	Discuss in brief the causes of failure of earth dams.	06
3.	a.	Explain the hydraulic design features of side channel spillways.	08
	b.	What is a stilling basin? List different types of stilling basins with neat sketches.	06
4.	a.	What is a surge tank? List the functions of surge tank.	07
	b.	What is the function of Anchor blocks? What are the forces to be taken to design the Anchor Blocks?	
5.	a.	Sketch the layout of power houses with components.	08
	b.	Draw neat sketches of vertical lift gate and tainter gate.	06
6.	a	What are the types of cross drainage works, list the same with neat sketches.	08
	b.	Sketch a offtake alignments mentioning its purpose.	06
7.	a.	Explain Bligh's creep theory with neat sketch. Also explain creep length and hydraulic gradient.	08
	b.	Explain the terms " safety against piping" and "safety against uplift pressure".	06
8.	a.	What are the components of diversion head works, sketch the layout of diversion head works.	10
	b.	Differentiate between barrage and weir.	04